

1. Specification section number:

Section 3.1, Table 3 — Possible responses to byte-level commands

2. Description of specification item:

Table 3 outlines the possible responses for every byte-level command. For the Read With Nak command, the potential outputs include Done, Arbitration Lost, and No Acknowledge.

3. Description of the bug:

Table 3 mistakenly includes No Acknowledge as a potential result for the Read With NAK command. According to the I²C protocol, the NAK that appears during a read cycle is issued by the Master to indicate that the transfer should stop, while the CMDR's NAK status is intended only for Slave-initiated non-acknowledgments. Because of this, a Slave is not capable of producing a NAK response during a Read With NAK transaction. The RTL already handles this scenario correctly - treating the state as unreachable—so the DUT behavior is unaffected. However, the specification needs to be updated to show that Done and Arbitration Lost are the only valid responses for this command.

4. Steps required to reproduce the bug:

- Write 0xC0 to CSR to enable core and interrupts
- Write bus ID to DPR
- Write CMD_SET_BUS (0x06) to CMDR, wait for DON
- Write CMD_START (0x04) to CMDR, wait for DON
- Write slave address with read bit (1) to DPR
- Write CMD_WRITE (0x01) to CMDR, wait for DON
- Write CMD_READ_NAK (0x03) to CMDR, wait for completion
- Observation of CMDR register - DON bit (bit 7) is set, NAK bit (bit 6) is never set regardless of slave behaviour

5. Waveform screenshot(s) showing inputs required to produce the bug and outputs displaying the bug:

